

Ardhendu Pramanik

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PROFESSIONAL SUMMARY

Frontend Engineer with strong full-stack capabilities and 2+ years building and shipping production-grade web applications in Next.js (App Router), React, and TypeScript. Owned features end-to-end on a live logistics platform — from frontend architecture to API integration — and independently designed, built, and deployed an AI-native, real-time collaborative system-design platform (multiplayer canvas + durable AI agents). Experienced in AI-assisted software engineering — Claude for architecture and implementation, CodeRabbit for automated review — under a Spec-Driven Development workflow. Strong in reusable component architecture, real-time/distributed state, performance optimization, and clean API design.

WORK EXPERIENCE

A2ZPORT · Software Engineer

Remote — Long Beach, CA · Oct 2023 – Sep 2025

- Engineered transport and container-tracking modules (Next.js App Router, React, TypeScript) handling **300+ active shipments daily** across dispatch and tracking workflows.
- Architected a library of **25+ reusable UI components**, cutting cross-module duplication by ~40% and accelerating new-feature delivery.
- Centralized state with Redux Toolkit, reducing unnecessary re-renders by ~35%; optimized API caching with React Query, improving average load time by ~20%.
- Wrote Zod schemas for runtime API request/response validation, reducing production data-shape errors after deployment.

The Internet Folks · Software Engineer

Bhopal, MP · Aug 2022 – Apr 2023

- Built production features in React / Next.js, translating Figma designs into pixel-accurate, responsive interfaces across multiple client products.
- Wrote unit & integration tests (Jest, React Testing Library) for critical UI components, reducing manual regression effort.
- Drove code-review standards that improved team-wide consistency in component patterns and API integration.

PROJECTS

Ghost AI — AI-Native Real-Time System-Design Platform · [Live](#)

Personal Project

Next.js 16 · React 19 · TypeScript · Liveblocks (CRDT) · React Flow · Trigger.dev · OpenRouter (LLM) · Prisma · PostgreSQL · Clerk · Vercel Blob

- Architected and shipped, solo and end-to-end, a multiplayer workspace where an **AI agent turns natural-language prompts into live architecture diagrams** and compiles the graph into a persisted Markdown technical spec.
- Built **real-time, conflict-free collaborative editing** with live presence and cursors on CRDT-backed Liveblocks + React Flow; AI agent emits a schema-validated action DSL with server-side sanitization for a valid graph regardless of model output.
- Moved all LLM work to **durable Trigger.dev background tasks** (retries, live progress) and enforced SHA-256 idempotency keys plus ownership checks at every mutation boundary.
- Engineered a two-tier storage model — PostgreSQL (Prisma) for metadata, Vercel Blob for artifacts — across **11 secured API routes** and a 22-component editor, delivered via an AI-assisted, Spec-Driven workflow.

Transport Management System — A2ZPORT · [Live](#)

Next.js · Supabase · Prisma · Redux · Tailwind · Zod

- Developed dispatch-workflow modules with dynamic, Zod-validated forms; modular component architecture scaled the team from 3 → 8 developers with minimal refactoring.
- Structured a scalable Prisma + Supabase data flow with production-grade handling of concurrency and data integrity across dispatch and tracking workflows.
- Implemented role-based access and reusable form/table primitives, accelerating delivery of new operational modules.

IVCA — The Internet Folks · [Live](#)

Next.js · Formik · Chakra UI · React Query · Jest

- Built responsive, accessible UI for the public platform of the **Indian Venture and Alternate Capital Association** — an industry body serving 250+ private equity, venture capital, and alternative-investment funds.
- Developed members, events, and research sections with optimized data-fetching (React Query) and Formik-driven forms; wrote integration tests covering key user flows.

SKILLS

Languages: TypeScript, JavaScript (ES2023+), SQL

Frontend: Next.js (App Router), React 19, Redux Toolkit, React Query, Tailwind CSS, shadcn/ui, React Flow

Backend & APIs: Node.js, Next.js API Routes / Server Actions, REST, Zod validation, Liveblocks (CRDT), Trigger.dev

AI Engineering: LLM app development, OpenRouter, prompt engineering, structured output, agentic workflows, AI-assisted dev (Claude)

Data & Infra: PostgreSQL, Prisma, MongoDB, Supabase, Clerk, Vercel, Git, GitHub Actions (CI/CD)

Practices: Spec-Driven Development, component architecture, performance optimization, system design

EDUCATION

B.Tech, Electronics & Communication (Lateral Entry) — CGPA 8.83/10

Aug 2019 – Jun 2022

Maulana Abul Kalam Azad University of Technology (MAKAUT)

Diploma, Electronics & Communication Engineering — CGPA 9.3/10

Aug 2016 – Jun 2019

WBSCTVESD

ACHIEVEMENTS & ACTIVITIES

- Built and deployed Ghost AI, an AI-native real-time collaboration platform, to deepen expertise in distributed systems, AI engineering, and end-to-end product development.
- Authored technical blogs on React performance optimization and TypeScript patterns.
- Continuously deepening expertise in AI-assisted engineering, agentic LLM workflows, Next.js App Router patterns, and testing (Jest/RTL).
- Regularly practice Data Structures & Algorithms to strengthen problem-solving and analytical skills.